Deep Learning Lab Practical – 6

Generative Models (VAE, GAN, CGAN, DCGAN)

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1. Objective

To design and implement generative models including Variational Autoencoder (VAE), Generative Adversarial Network (GAN), Conditional GAN (CGAN), and Deep Convolutional GAN (DCGAN) for image generation using the FashionMNIST dataset.

2. Dataset

FashionMNIST dataset was used, consisting of grayscale images of size 28×28 with 10 classes such as shirts, shoes, and bags.

3. Models Implemented

3.1 Variational Autoencoder (VAE)

Encoder-decoder architecture
Uses latent space representation
Loss: Reconstruction + KL Divergence

3.2 Generative Adversarial Network (GAN)

Generator and Discriminator
Trained using adversarial learning

3.3 Conditional GAN (CGAN)

Adds label information to control output

3.4 Deep Convolutional GAN (DCGAN)

Uses convolutional layers for better image quality

4. Results

VAE:
- Successfully reconstructed images
- Generated new images from latent space
- Output is slightly blurry

GAN:
- Generated sharper images compared to VAE
- Training shows instability in loss curves

(Add screenshots of outputs here)

5. Comparison

VAE – Stable but blurry
GAN – Sharp but unstable
CGAN – Controlled generation
DCGAN – Best image quality

6. Conclusion

VAE provides stable training but produces blurry images. GAN generates sharper images but is unstable. CGAN allows controlled generation using labels, while DCGAN improves image quality using convolutional layers.